

SMART INDIA HACKATHON 2026



Sirens of Summer

Problem Statement ID - SIH26083

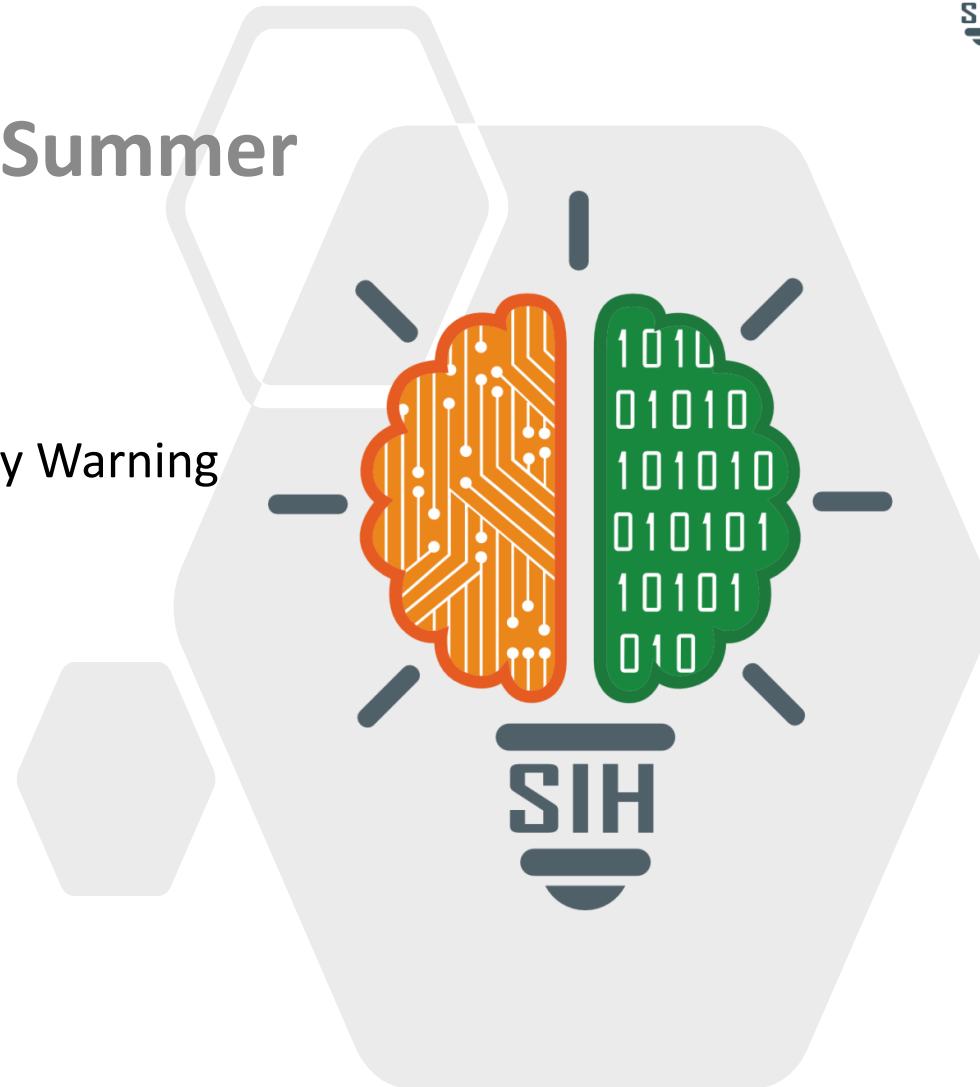
Problem Statement - Extreme Heatwave Early Warning
& Human Thermal Stress Index

Theme - Disaster Management

PS Category - Software

Team ID - [Your Team ID]

Team Name - Sirens OF SUMMER



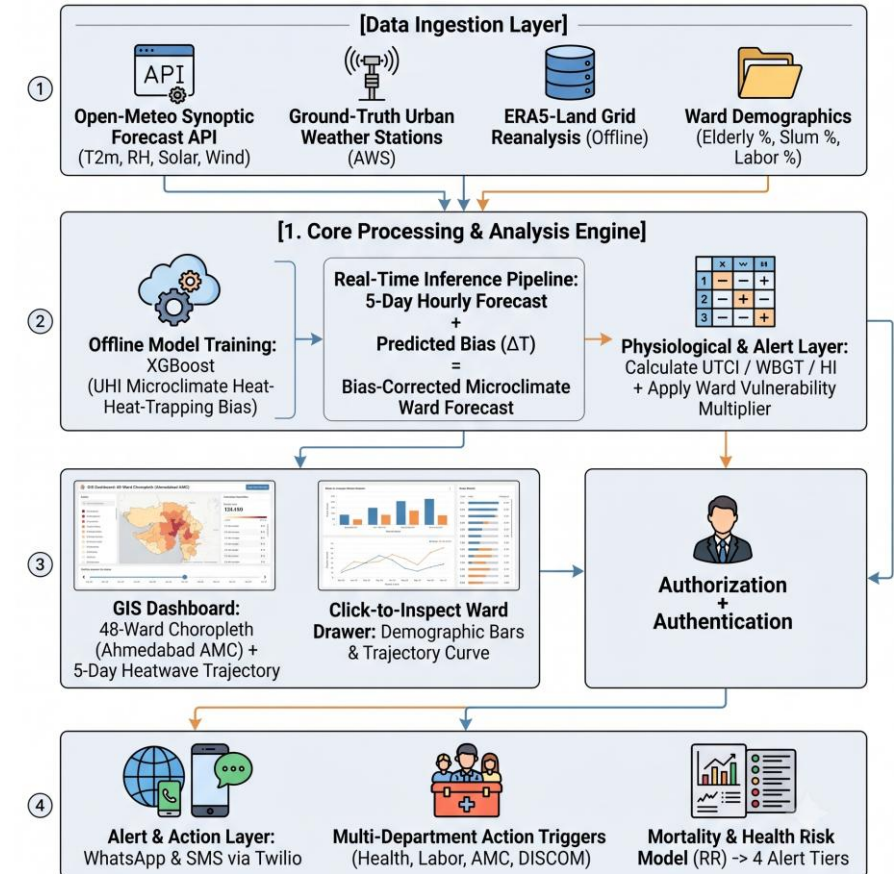
❖ Proposed Solution :-

- AI-powered, local early warning at ward level for Indian metropolitan cities.
- 5-day bias-corrected microclimate forecasts plus UTCI, WBGT and Heat Index.
- Ward-level mortality and health risk scoring with automated department action triggers

❖ Why is it different?

- Moves from predicting temperature to actionable intelligence: UTCI, alerts and recommended action for each ward
- Combines hazard with elderly population, outdoor workers, slum housing and green-cover deficit
- Dashboard for ward inspector and alert view for all the wards
- Explainable AI shows why a ward is at risk

Extreme Heatwave Early Warning & Human Thermal Stress Index: System Architecture & User Pipeline



SIH-26083 | MoES / NCMRWF

- **Data Ingestion (Archive & Live)**

Hourly weather feeds via Open-Meteo API (T2m, RH, solar radiation, wind speed).

- **Diurnal & Microclimate Preprocessing**

Extract diurnal sin/cos harmonics, 24h thermal lag, and localized UHI biases (ΔT).

- **XGBoost Bias-Correction Engine**

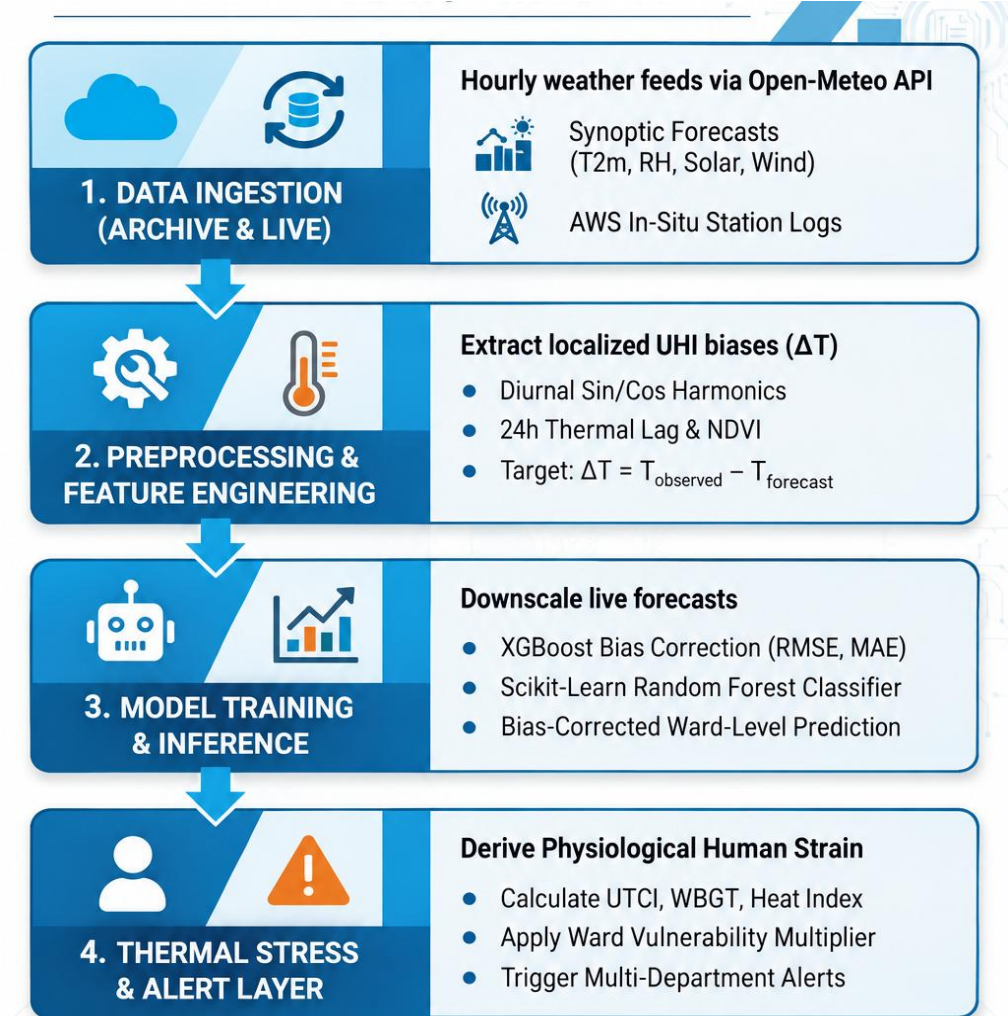
Trained regressor corrects live 5-day synoptic forecast to ward-specific temperatures.

- **Thermal Stress (UTCI) & Alert Layer**

Compute physiological human strain index and classify risk into actionable tiers.

- **Interactive Folium & Streamlit UI**

Real-time 15-zone Delhi-NCR choropleth polygon visualization and early warning dispatch.



Analysis of Idea Feasibility

Technical Viability: Utilizing Open-Meteo APIs and lightweight XGBoost regression ensures the system requires low computational overhead, allowing rapid real-time 5-day inference on standard hardware.

Algorithmic Standardization: Computing complex human thermal stress is highly feasible using established, open-source biometeorology libraries like pythermalcomfort to calculate ISO 7243 WBGT and UTCI efficiently.

Deployment Readiness: The integration of Streamlit and Folium allows for fast, modular GIS dashboard deployment, while Twilio APIs provide a ready-made, highly accessible dissemination channel for WhatsApp and SMS.

Potential Challenges & Risks

- **Ground-Truth Data Sparsity:** In-situ Automatic Weather Stations (AWS) are sparsely distributed in many regions and prone to sensor downtime, making continuous localized validation difficult.
- **Urban Heat Island (UHI) Resolution Limits:** Coarse synoptic forecasts often fail to capture block-level UHI effects, where urban temperatures can spike significantly higher than surrounding rural areas by 1–7 ° C due to dense built environments.
- **Alert Fatigue:** Sending excessive, generic heat warnings via mobile channels can cause desensitization, leading vulnerable populations or administrative bodies to ignore critical, life-saving alerts.

Strategies for Overcoming Challenges

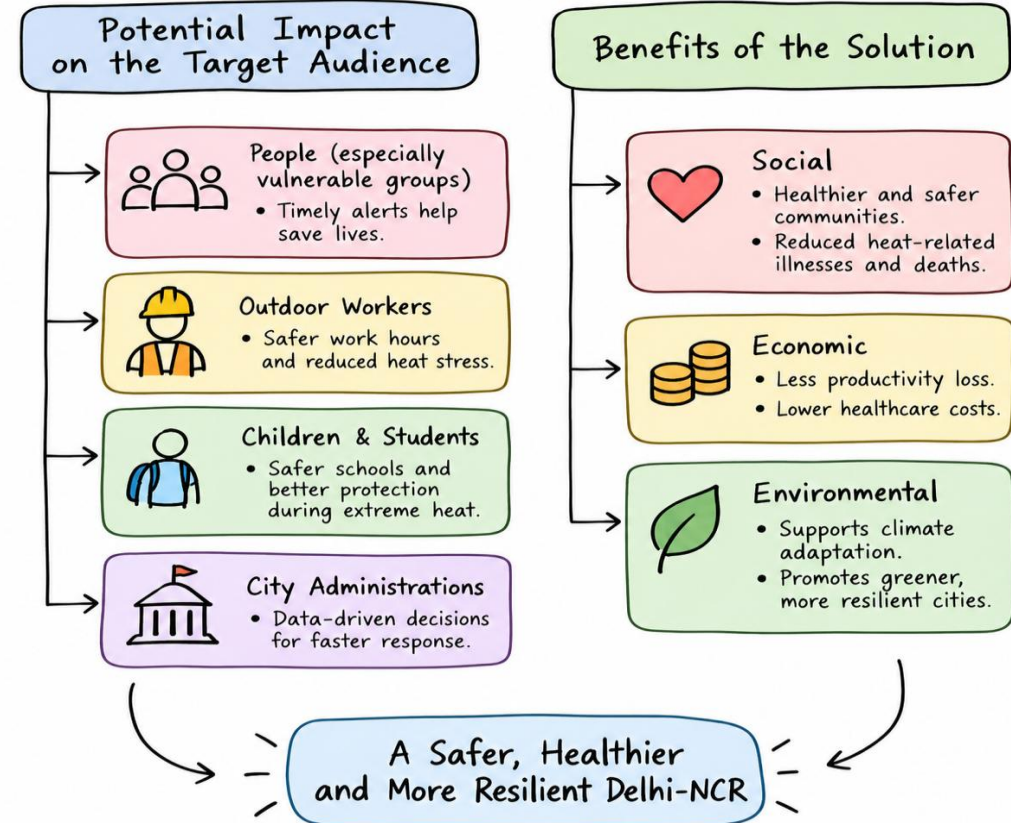
- **AI Bias-Correction Engine:** Train the XGBoost model using spatial proxies—like NDVI vegetation deficits, slum density, and 24-hour thermal lags—to predict and correct microclimate biases (ΔT) at the ward level without requiring a dense grid of physical sensors.
- **Synthetic Fallback Architecture:** If live AWS data drops, the pipeline is engineered to automatically fall back to historical ERA5-Land reanalysis grids and synthetic mean-bias baselines to guarantee continuous, zero-downtime forecasting.
- **Tiered & Targeted Dispatch:** Use the calibrated Tri-Factor Risk Model (Hazard + Exposure+Vulnerability) to restrict direct Twilio alerts to high-risk Orange/Red tiers, sending specific, actionable directives to distinct departments rather than generic mass broadcasts.

Potential Impact on the Target Audience

Our system provides localized, ward-level heat-risk alerts based on human thermal stress rather than temperature alone. It helps vulnerable groups such as elderly people, outdoor workers, children, and low-income communities take timely preventive action, while enabling healthcare services and city authorities to prepare resources and respond faster.

Benefits of the Solution

The solution delivers social, economic, and environmental benefits by reducing heat-related illness and mortality, minimizing productivity losses and healthcare costs, and improving emergency preparedness. It also supports climate adaptation, targeted urban planning, and long-term development of more heat-resilient cities.



Data Sources & APIs :-

- Open-Meteo API — <https://open-meteo.com/en/docs>
- Copernicus ERA5-Land — <https://cds.climate.copernicus.eu/datasets/reanalysis-era5-land>
- pythermalcomfort — <https://pythermalcomfort.readthedocs.io/en/latest/>
- NDMA – Heatwave Guidelines / Heatwave Information — <https://ndma.gov.in/>
- Census of India — <https://censusindia.gov.in/>
- Census Data / Tables — <https://censusindia.gov.in/census.website/en/data>
- IMD – Heatwave Guidance & Bulletins — https://mausam.imd.gov.in/responsive/heatwave_guidance.php
- IMD – District-wise Heatwave Warnings — <https://mausam.imd.gov.in/responsive/districtWiseHeatwaveWarnings.php>
- NOAA – Weather — <https://www.noaa.gov/weather>